

Erika Coreth

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Professional Summary

Master's student in Computer Science (Concentration in Software Engineering) with experience in Python, C++, Java, and SQL. Strong foundation in healthcare and biomedical data analysis, with proven ability to design, test, and implement reliable applications and databases.

Education

Merrimack College <i>Master of Science in Computer Science, Concentration in Software Engineering</i>	North Andover, MA Expected: March 2026
University of Connecticut <i>Master of Science in Surgical Neurophysiology</i>	Storrs, CT
Monmouth University <i>Bachelor of Science in Health Studies, Minor in Biology and Chemistry</i>	West Long Branch, NJ

Technical Skills

Languages: Python, C++, SQL, Java

Developer Tools: GitHub, VS Code, Visual Studio, PyCharm, IDLE, MySQLWorkbench, MATLAB, HAPPE+ER, EclipseIDE

Work Experience

EEG Data Processing Assistant - Part-Time

Merrimack College North Andover, MA
Supervisor Dr. Cremonese-Caira June 2025- Aug 2025

- Executed preprocessing and analysis of 21 electroencephalography (EEG) and event-related potential (ERP) subject files, enabling inclusion of data in research studies.
- Developed and modified MATLAB scripts to resolve compatibility issues and automated data imports, enabling successful processing of EEG subject files and reducing manual processing time by ~30%.
- Applied signal processing techniques (artifact rejection, filtering, re-referencing) to clean preprocessed EEG data, improving signal quality and retaining ~85-90% of usable trials.
- Generated data visualizations and statistical summaries across 7 datasets, accelerating reporting workflows and supporting research publication readiness.
- Resolved recurring MATLAB/HAPPE+ER workflow errors, restoring full pipeline functionality and enabling 100% dataset processing without crashes.
- Strengthened technical expertise in MATLAB scripting, biomedical signal analysis, and pipeline-based data automation.

United Medical Monitoring

Surgical Neurophysiologist NJ/PA
June 2023-Present

- Monitored patients' nervous systems during surgery to help prevent complications and ensure patient safety.
- Utilized various techniques to record and measure electrical activity in the brain, nerves, muscles, and spinal cord.

Projects

Philadelphia Eagles Database Application (Full-Stack Project, SQL + Python)

- Designed and implemented a MySQL database with multiple normalized tables, foreign keys, views, and stored procedures.
- Developed a three-tier MVC application in Python:
 - DAL (Data Access Layer) - handled SQL queries and stored procedures
 - BLL (Business Logic Layer) - enforced logic and validation
 - GUI (PySimpleGUI Front-End) - allowed users to add, edit, delete players, query game schedules, and export reports to CSV files.
- Created SQL views to display player and team statistics in user-friendly formats.
- Integrated exception handling, formatted outputs, and dynamic user input validation.

Lyric Text Editor (Python Project)

- Built a custom text editor in Python to analyze and manipulate song lyrics.
- Implemented functionality for:
 - Word frequency analysis and most-frequent word detection.
 - Replacing, adding, and deleting text.
 - Highlighting specific words in the lyrics file.
- Automated file operations to save edits directly to the lyrics file.
- Practiced file handling, string manipulation, and text-processing algorithms in Python.